SECUREVAULT

Project components and estimated prices | India | 20 October 2026

Prototype budget

Estimated hardware total: Rs 2,014

Plan for roughly Rs 1,600-3,000 overall. Prices vary by seller, quality, location, and delivery.

Hardware

Component	Qty	Budget	Notes
ESP32 DevKit V1	1	Rs 500	Main controller with Wi-Fi; generic board estimate
RC522 RFID reader	1	Rs 99	Use at 3.3 V
RFID cards / tags	2	Rs 50	One enrolled tag plus one spare
4x4 membrane keypad	1	Rs 85	PIN and OTP entry
SG90 / MG90S servo	1	Rs 150	Small demo latch; MG90S may cost more
PIR / tamper sensor	1	Rs 69	HC-SR501 class sensor
Buzzer	1	Rs 29	Active buzzer estimate
LEDs	2	Rs 10	Status indicators
220-ohm resistors	2	Rs 5	For the LEDs
Breadboard	1	Rs 89	Prototype wiring
Male-male jumper wires	1 pack	Rs 59	Pack estimate
Male-female jumper wires	1 pack	Rs 59	Pack estimate
USB cable for ESP32	1	Rs 80	May already be included with board
External regulated 5 V supply	1	Rs 250	Allow separate servo supply with shared ground
Small locker / project box	1	Rs 250	Simple wood/acrylic enclosure estimate
Servo mounting bracket	1	Rs 80	Generic bracket estimate
Mechanical latch	1	Rs 150	Simple demo latch estimate
Estimated hardware subtotal		Rs 2,014	Before shipping and taxes where applicable

Software and services

Item	Estimated cost	Notes
Arduino IDE + ESP32 board support	Free	Firmware development and upload
Python 3 + Flask	Free	OTP backend on an existing laptop
VS Code or another code editor	Free	Optional editor
Web browser / phone dashboard	Free	Local network demo; assumes devices already available
OTP delivery service / cloud hosting	Not included	Optional; cost depends on provider and delivery method

Cost assumptions

The subtotal uses a budget build with a generic ESP32 board and basic mechanical parts. It is an estimate, not a live quote. It excludes shipping, tools, spare parts, and any paid SMS/email OTP service or cloud server. If you already have the laptop, phone, USB cable, or Wi-Fi, reuse them to reduce the cash outlay. The quoted lock is a tabletop/college demonstration prototype; an SG90 servo and simple latch are not a substitute for a rated security lock.

Build and buying notes

- Prototype on the breadboard first, and buy or build the final box after the servo and latch move reliably.
- Power the RC522 from 3.3 V. Do not power the servo from an ESP32 GPIO; use the separate regulated 5 V supply and connect its ground to ESP32 ground.
- For a no-subscription demonstration, run the Flask backend on an existing laptop and keep the ESP32, laptop, and phone on the same Wi-Fi. SMS delivery and remote cloud hosting are optional costs.

Useful buying references

RC522 reader: Robu.in - Rs 99 listing | ESP32 examples: Robu.in ESP32 board listings

PIR sensor: Robu.in - HC-SR501 listing around Rs 69 | Keypad: TecnoMate - Rs 87 listing

Servo: Tech Depot India - SG90 listing around Rs 151 | Small parts: CaressTech component price examples

Listings checked on 20 October 2026. Prices and stock can change; check delivery charges and compatibility before ordering.